

# WENDELL MCGOWAN III

## Geospatial Software Engineer

Colorado Springs, CO • [wendell.d.mcgowan@gmail.com](mailto:wendell.d.mcgowan@gmail.com) • (702) 902-8370

[linkedin.com/in/wendell-mcgowan-5798a328b](https://www.linkedin.com/in/wendell-mcgowan-5798a328b) • [github.com/TheWendarr](https://github.com/TheWendarr) • [Portfolio Webpage](#)

Active TS/SCI • U.S. Army Staff Sergeant, transitioning (SkillBridge-eligible mid-2027; available September 2027)

## SUMMARY

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Geospatial software engineer with 8+ years of U.S. Army geospatial engineering experience and a strong spatial-data development portfolio. Builds production Python and Rust tooling for terrain analysis, raster/vector processing, and geospatial pipelines including an open-source H3-based cross-country mobility model that modernizes legacy Army terrain-analysis workflows. Comfortable across the stack: ArcGIS Pro/Enterprise and QGIS, GDAL-based data engineering, Flask web applications, and self-managed Linux infrastructure. Proven record leading technical teams and delivering standards-compliant products under operational pressure.

## TECHNICAL SKILLS

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**Languages:** Python (advanced) • Rust • SQL • Bash

**Geospatial:** ArcGIS Pro • ArcGIS Enterprise / Experience Builder • QGIS • arcpy • H3 tessellation • GDAL • rasterio • geopandas • shapely • exactextract • pyogrio • Esri file geodatabases • branch versioning

**Web & Data:** Flask • Polars • SQLite • REST APIs • OGC services

**Systems:** Linux (Fedora / Debian) • Git & GitHub • Docker / Podman • networking (subnetting, DNS, VPN)

## PROFESSIONAL EXPERIENCE

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### Geospatial Engineer (12Y) — Staff Sergeant (E-6)

U.S. Army, 4th Infantry Division GEOINT Cell - Fort Carson, CO November 2024 - Present

- Serve as Database Manager for the division GEOINT Cell, administering spatial databases and geospatial data infrastructure supporting GEOINT production for a 17,000-soldier division
- Produce cartographic-grade maps and geospatial intelligence products conforming to NSG standards, directly supporting division-level planning, operational assessments, and time-sensitive intelligence requirements
- Analyze complex multi-source geospatial datasets — integrating elevation, slope, vegetation, and soil rasters — to derive terrain intelligence products used in operational decision-making
- Engineer repeatable Python (arcpy) workflows that automate raster processing and product generation, reducing manual production time for recurring terrain products
- Coordinate geospatial data requirements across staff sections and higher echelons; mentor and train junior Geospatial Engineers in GIS software, GEOINT workflows, and data management

### Geospatial Engineer / Motor Transport Operator - E-5/E-6

Nevada Army National Guard — Las Vegas, NV December 2017 – November 2024

- Produced GEOINT and cartographic products supporting unit training exercises and state-level emergency response operations, consistently meeting quality and timeliness standards
- Operated and maintained GIS workstations, geospatial databases, and field survey equipment (Total Station) in garrison and field environments
- Led and supervised teams of up to 8 soldiers across geospatial and logistics missions under time-constrained and austere conditions
- Managed sensitive geospatial data in strict accordance with classification requirements and data-handling protocols

## TECHNICAL PROJECTS

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### Hex Surface Model - Cross-Country Mobility Pipeline - Python • open source (MIT) • degree capstone

[github.com/TheWendarr/hex-surface-model](https://github.com/TheWendarr/hex-surface-model)

- Designed and built an automated terrain-analysis pipeline that tessellates terrain into H3 hexagonal cells, computes multi-factor surface costs from elevation, slope, land cover, and hydrology rasters, and routes optimal paths per vehicle mobility profile via hierarchical A\* search
- Implemented multi-factor raster zonal statistics with exactextract and a modular geospatial I/O layer (geopandas, pyogrio, shapely); achieved a ~2.2× pipeline speedup through profiling and platform optimization

- Modernizes the Army's manual MCOO terrain-analysis workflow into a repeatable, parameterized software pipeline; documented in an APA 7 research paper

**Runner's Dashboard** - Python • Flask [github.com/TheWendarr/runners-dashboard](https://github.com/TheWendarr/runners-dashboard)

- Built a privacy-first web dashboard that parses Garmin FIT binary files entirely in memory — no cloud upload, no data persisted — and renders interactive pace, distance, and heart-rate analytics with imperial/metric support
- Re-architected a 3-script CLI pipeline into a single cohesive Flask application with a structured parsing layer and chart API

**Portfolio Site** - [Portfolio Website \(Hosted by Github\)](#) - Vanilla JS • Canvas • Leaflet

- Designed and built an interactive portfolio with a custom canvas-rendered H3 hexagonal cost surface over a Leaflet basemap, a data-driven project portfolio

## EDUCATION

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**B.A.S., Geospatial Intelligence & Analysis**

*Delta State University - Cleveland, MS* Expected December 2026

**Relevant coursework:** Programming GIS • Internet, GIS & Spatial Databases • Geospatial Mathematics & Algorithms • Computer Programming Fundamentals • Geospatial Analysis • Structured Analytic Techniques • Imaging Science

**Capstone:** Hex Surface Model (see Technical Projects)

## CERTIFICATIONS

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- GPC-F
- CompTIA Network+
- USGIF GEOINT Essentials Certificate